WarpVox: Observation-Guided Deformable Voxel Maps for Efficient TSDF Correction after Loop Closure

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***Abstract*— Loop closure retroactively changes sensor poses after their range observations have been fused, leaving a TSDF truncated signed distance field (TSDF) inconsistent with the corrected trajectory. Full reintegration restores consistency by replaying the complete measurement history, at a prohibitive runtime cost that grows with the session length and is paid again at every correction. WarpVox instead corrects the active map without volumetric replay: projecting the current surface into keyframes re-establishes measurement – surface correspondences; corrected poses re-express the matched samples as surface targets, a single deformation propagates the targets, and the TSDF is rebuilt around the deformed surface. The result is a native Voxblox TSDF whose zero set follows the corrected surface while its volumetric support, sign, and weights primarily follow the source volume; it replaces the active map, allowing fusion and subsequent loop-closure corrections to continue from the updated state. Across 51 loop closures on Kimera-Multi and KITTI, WarpVox stays within 0.32 percentage points of full reintegration at in dataset-mean F@25 while achieving a $48\times$ mean update-time speedup on the RGB-D sequences. Compared with Kimera-PGMO, a deformation-graph mesh optimization method, WarpVox achieves comparable corrected-surface accuracy at lower costcorrects the surface more accurately at F@25/F@50 and in Chamfer distance, in less than half the time. A single-sequence case study extends the same targets to Gaussian centres.**

I. Introduction

Dense 3D maps support navigation, manipulation, and scene understanding in robotics and augmented reality. TSDF Truncated signed distance field (TSDF) methods [1], [2] integrate range observations incrementally, while sparse volume structures [3]–[5] make large-scale onboard mapping practical. These systems accept poses from an external SLAM simultaneous localization and mapping (SLAM) front-end. Loop closure and pose-graph optimization [6], [7] retroactively correct the trajectory, leaving the TSDF in the old world geometryand therefore , inconsistent with the revised poses (Fig. 1).

The standard remedy is to discard the map and reintegrate all N range frames with corrected poses [8], [9]. Its $O(N|D|)$ cost, for $|D|$ measurements per frame, is paid again after every global correction and can reach minutes on long sessions. A complete de-integration/reintegration cycle removes and re-adds every affected frame. Because As a global loop closure can affect nearly the complete whole history, this approaches two full replay passes. Bounded schedules such as FlashFusion [10] update only a limited subset online, temporarily mixing corrected and stale contributions. We therefore target long-running online robotic mapping: consume an already fused TSDF and a revised trajectory, and return a fusion-ready TSDF without requiring access to the complete range history.

Submap methods such as Voxgraph [11] avoid full replay by optimizing rigid SDF submaps, but require the map to be partitioned during fusion and leave intra-submap drift unresolved. Other representative approaches change the representationinstead: non-rigid surfel SLAM [12] corrects its own tracking state and does not update an externally maintained TSDF, while neural implicit maps [13]–[16] require representation-specific re-optimization. Each is a sound design; none returns a corrected TSDF to a mapper already maintaining one.

Our starting point is an invariance: a calibrated range sample expressed in its sensor frame is unchanged by a later world-frame pose correction, although its relation to the current map surface is not. WarpVox therefore uses keyframe range measurements to rebuild projective measurement – surface correspondences at each correction event. Re-expressing the matched samples with corrected poses yields corrected world-space surface targets by projection alone. The expensive volumetric update therefore scales with the current map, while repeated volumetric replay of the complete measurement history is eliminated.

WarpVox combines four stages: (i) projectively re-associate keyframe range measurements with the current pre-correction surface, (ii) compute robust targets from the corrected poses, (iii) propagate sparse displacements through a single ARAP as-rigid-as-possible (ARAP) solve [17] on a QEM quadric-error-metric (QEM) proxy, and (iv) reconstruct a TSDF whose zero set follows the warped surface while observed support, sign, and weight primarily follow the source volume. At each loop closure, the corrected volume replaces the inconsistent active TSDF; Voxblox then resumes fusion from the updated state until a later trajectory revision triggers the same correction procedure again.

- An observation-derived target construction that turns keyframe range measurements and a revised trajectory into corrected surface targets by projection, without replaying volumetric integrationtraces fusion backwards: the keyframe measurements behind each surface point are recovered and screened under the old poses and combined under the corrected poses, without volumetric replay.
- A source-supported TSDF reconstruction that combines the warped zero set with source-derived support, sign, integration weights, and free space, returning a native Voxblox volume that replaces the active TSDF and supports continued fusion and later corrections.
- An evaluation of over 51 RGB-D and LiDAR correction loop closures against reintegration and mapping baselinesover 51 loop-closure cases, together with a separate comparison of final-state surface correction , with comparisons against

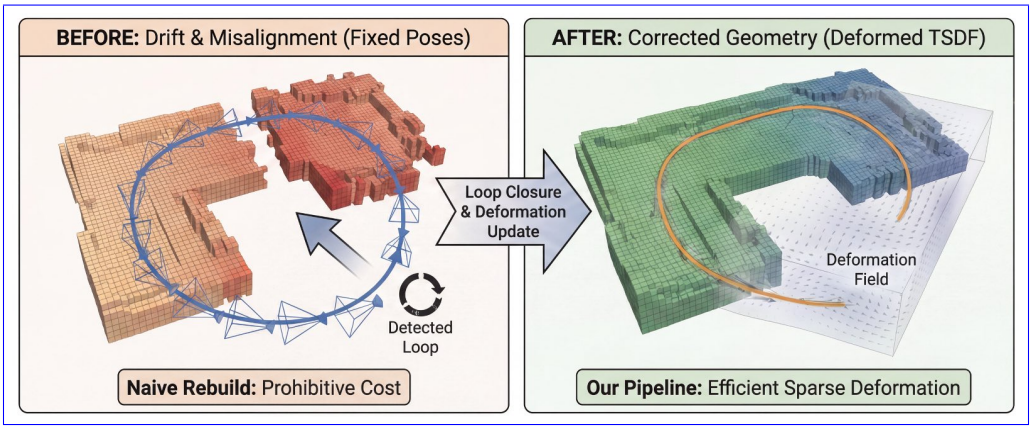


Fig. 1. **WarpVox corrects volumetric TSDF maps after loop closure without re-fusing raw data.** A detected loop revises the trajectory, leaving the fused map inconsistent with the revised poses. Reintegration replays every frame; WarpVox deforms the current surface and reconstructs a corrected TSDF.

Kimera-PGMO on ten complete bagsand ElasticFusion and repeated correction with continued fusion.

II. Related Work

Volumetric mapping. TSDF fusion [1], [2] and sparse volume structures [3]–[5] enable incremental dense mapping from externally supplied poses. These systems normally assume that the input trajectory remains fixed and do not expose a mechanism for revising a persistent volume after a pose-graph backend changes historical poses. Wavemap [18] achieves state-of-the-art memory and computational efficiency through wavelet-compressed multi-resolution occupancy mapping; as an occupancy representation, it likewise rebuilds from measurements after a trajectory revision.

Dense-map correction after loop closure. Kintinuous [19] deforms point-cloud slices, while ElasticFusion [12] deforms a surfel map within its tracking system. Maier et al. [20] correct a voxel-hashed SDF after pose-graph updates by reintegrating fused RGB-D keyframes; WarpVox uses keyframe measurements only to construct deformation targets and performs no volumetric replay. Voxgraph [11] optimizes rigid TSDF submaps, making correction cheap but requiring a partitioned volume and leaving intra-submap drift. Kimera-PGMO [9] jointly optimizes poses and a timestamped mesh, producing only a corrected surface, not a volume.

Surface deformation. ARAP [17] and embedded deformation [21] preserve local shape while satisfying sparse controls. DynamicFusion [22], KillingFusion [23], and VolumeDeform [24] estimate continuous non-rigid scene motion. WarpVox addresses a static scene with a discrete trajectory revision: corrected poses provide controls, propagated by a single QEM-proxy surface deformation.

Neural and Gaussian maps. Neural SLAM systems [13], [14], [25] optimize implicit scene representations; Loopy-SLAM [15] revisits that optimization after loop closure, while Neural Graph Map [16] anchors local fields to pose-graph nodes. SplaTAM [26] maps with 3D Gaussians [27], and LoopSplat [28] aligns Gaussian submaps. These representations do not generally provide the explicit volumetric state required by TSDF pipelines. WarpVox therefore targets such pipelines; its Gaussian experiment evaluates only the transferability of observation-derived targets.

III. Method

Assumptions. Following the local-consistency assumption of WarpVox corrects an existing truncated signed distance field (TSDF) after loop closure, using stored keyframe measurements and the externally corrected trajectory. TSDF fusion forms each surface point from the range measurements that observed it, placed by their frame poses; a loop closure revises those poses but leaves the measurements valid in their sensor frames. WarpVox traces the fusion backwards: Stage 1 recovers, for each surface vertex, the keyframe measurements that formed it; Stage 2 places them under the corrected poses as observation-derived targets; Stage 3 propagates these targets over the surface; and Stage 4 combines the corrected geometry with source support, sign, weights, and free space into a native TSDF for continued fusion.

A. Problem and Correction Inputs

Following Voxgraph [11], we assume that the pre-correction TSDF remains is assumed locally valid: surface geometry and connectivity remain stay coherent in local neighbourhoods despite global inconsistency. Severely , and severely noisy or topologically corrupted source maps are outside our scope.

B. *Problem and Correction Inputs*

For any point For $x \in \mathbb{R}^3$, let $\widetilde{x} = [x^\top, 1]^\top$ denote its denotes homogeneous coordinates. Let $T^-(t), T^+(t) \in SE(3)$ map the sensor frame at time t to the world frame before and after trajectory the correction. The map built with T^- consists of a TSDF Φ^-, its weight field w^-, is a Voxblox TSDF [4]: a regular grid of voxel size s, and truncation distance τ. During online fusion, Voxblox continuously maintains allocated sparsely in blocks. Each voxel centre x stores a truncated signed distance $\Phi^-(x) \in [-\tau, \tau]$, positive on the sensor side, and an integration weight $w^-(x) \geq 0$, zero if unallocated. A voxel is *supported* if its weight is positive; the supported voxels form the observed support Ω^-. At a correction,

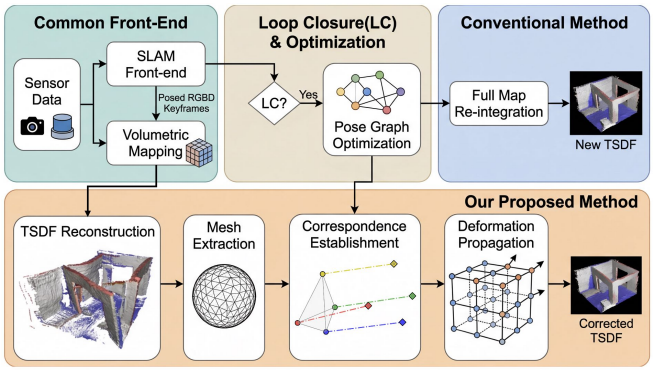


Fig. 2. WarpVox extracts the current source surface from the fused TSDF, establishes projective correspondences with keyframe measurements, propagates the resulting targets through one proxy deformation ARAP solve per correction, and reconstructs a source-supported TSDF for continued fusion.

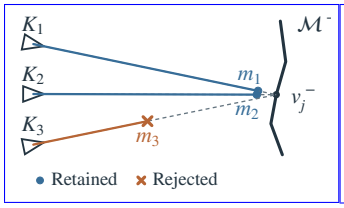


Fig. 3. Stage 1: association of keyframe observations.

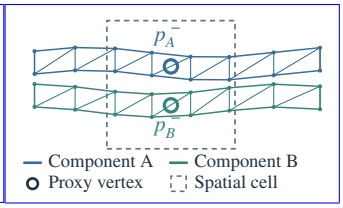


Fig. 4. Stage 3: one proxy vertex per (cell, component) pair.

the Marching-Cubes surface$\mathcal{M}^- = (V^-, F)$ of the active TSDF [29] . Upon trajectory correction, this mesh [29] that Voxblox maintains during fusion becomes the source surface . Let $\mathcal{A}$ denote the active TSDF cells contributing triangles to $\mathcal{M}^-$. WarpVox stores the timestamped , modality-native range images of SLAM keyframes,

$$\mathcal{C}_{\mathrm{KF}} = \{k_i = (t_i, D_i) : i \in \mathrm{KF}\},$$

where KF indexes the retained SLAM keyframes and D_i is an RGB-D depth image or a spherical LiDAR range image . Together with T^-, these measurements form the observation input to Stage 1. The measurements remain available across correction events. $\mathcal{M}^- = (V^-, F)$: vertex positions $V^- = \{v_j^-\}$, triangle indices F, and area-weighted vertex normals n_j^-. WarpVox stores one timestamped range image per keyframe (KF) in its sensor frame, $\mathcal{C}_{\mathrm{KF}} = \{(t_i, D_i)\}$: the depth image of each SLAM keyframe (RGB-D) or the spherical range image of each posed scan (LiDAR). Fig. 2 summarizes the four stages, Table I lists all parameters, and Figs. 3 – 5 illustrate association, clustering, and reconstructed support.

The observation records and warped mesh are intermediate; WarpVox returns a native Voxblox volume (Φ^+, w^+).

B. Stage 1: Correspondence Establishment

Stage 1 establishes projective correspondences between the current source surface and the keyframe range recovers the measurements that formed the surface by reversing the projective fusion rule, associating source vertices with geometrically consistent keyframe measurements under the pre-correction trajectoryposes T^-. For each queried source vertex v_j^- with normal

TABLE I

WarpVox parameters, one setting per modality for all datasets (s: voxel size).

Quantity	RGB-D	LiDAR
Stage 1 (Sec. III-B)		
KF; range image D_i	SLAM KF; depth image	scan; 64×2048 image
Queried vertices; index radius	all; 7.2 – 7.4 m	1 per 0.5 m cell; 25 m
Valid range	0.1 – 5 m	0.5 – 25 m
Gate $g_0 + g_1 z_{rj}^-$	$0.08 + 0.01 z_{rj}^-$ m	$0.25 + 0.003 z_{rj}^-$ m
σ_g=$0.6 g_0$ / σ_z / z_0, Eq. (2)	0.048 / 0.08 / 8 m	0.15 / – / 40 m
$\alpha_{\min}$ / $n_{\max}$	0.05 / 15	– / 15 of first 30
Stage 2 (Sec. III-C)		
Pose $\widehat{T}_r^+$	interp. (gap $\leq 3\times$ median)	scan pose
ζ_m / ϵ_m / κ	3 m / 0.02 m / 2.5	20 m / 0.05 m / 2.5
Stage 3 (Sec. III-D, Alg. 1), both		
N_{f} / k / $n_{\min}$ / γ / $\sigma_{\min}$	100 faces / 20 / 16 / 1.5 / 0.05 m	
$P_{\max}$ / ARAP iterations	60,000, proxy if $\lvert V^-\rvert > 1.3P_{\max}$ / 20	
Welsch rounds / c_{g} / min. nbrs.	4 / 4.685 / ≥ 6	
τ_{e} / τ_{A} / τ_{asp} / $\ell_{\min}$	3 / 6 / 20 / 0.5 m	
n_{rep} / Jacobi steps per pass	3 / 80 (both at most)	
Stage 4 (Sec. III-E), both		
τ / ρ / $w_{\max}$	$4s$ / $1.5s$ / 10^4	
Source transport cell width	τ	

n_j^-, a spatial index over sensor positions , a k-d tree (RGB-D) or grid (LiDAR) over the sensor centres retrieves candidate keyframes that may observe it.

Let r index a candidate measurement within the index radius. For candidate r, with timestamp t_r and range image D_r. The source vertex is transformed into the corresponding , the vertex is mapped into its sensor frame and sampledin the measurement domain:

$$\hat{v}_{rj}^- = \left[[T^-(t_r)]^{-1}\widetilde{v}_j^-\right]_{1:3},$$
$$m_{rj} = \mathcal{S}_m\left(D_r, \Pi_m(\hat{v}_{rj}^-)\right).$$

:

$$\hat{\mathrm{v}}_{\mathrm{rj}}^- = \left[[\mathrm{T}^-(\mathrm{t_r})]^{-1}\widetilde{\mathrm{v}}_{\mathrm{j}}^-\right]_{1:3}, \qquad \mathrm{m_{rj}} = \mathcal{S}_{\mathrm{m}}\left(\mathrm{D_r}, \Pi_{\mathrm{m}}(\hat{\mathrm{v}}_{\mathrm{rj}}^-)\right). \quad (1)$$

Here For modality m, Π_m is the pinhole projection for (RGB-Dor the spherical projection for LiDAR. For RGB-D, $\mathcal{S}_m$ samples the projected depth) or spherical (LiDAR) projection, and back-projects it to the sensor frame. For LiDAR, it selects the closest valid return in a local neighbourhood of the projected spherical-image location$\mathcal{S}_m$ returns one sensor-frame point $m_{rj} \in \mathbb{R}^3$, or none: the depth bilinearly interpolated from at least two valid pixels and back-projected (RGB-D), or the return closest to $\hat{v}_{rj}^-$ in a 3×3 window (LiDAR).

Association quality is evaluated from the discrepancy between the projected source vertex and the sampled measurement. For Let z_{rj}^- and z_{rj}^m be the depth (RGB-D, this includes the three-dimensional point discrepancy and projective depth difference. For LiDAR , the local spherical-image lookup evaluates radial-range and three-dimensional consistency. Viewing incidence is evaluated in the pre-correction world frame from) or range (LiDAR) of $\hat{v}_{rj}^-$ and m_{rj}, $\delta_{rj} = \|m_{rj} - \hat{v}_{rj}^-\|$, and ϕ_{rj} the angle between n_j^- and the sensor-to-vertex direction.

These cues determine whether a the viewing ray. A sample must pass the gate $\delta_{rj} \leq g_0 + g_1 z_{rj}^-$, which for LiDAR also bounds $|z_{rj}^m - z_{rj}^-|$, and receives the reliability

$$\alpha_{\mathrm{rj}} = \frac{|\cos\phi_{\mathrm{rj}}|}{1 + \mathrm{z_{rj}^-}/\mathrm{z_0}}\, \mathrm{e}^{-\delta_{\mathrm{rj}}/\sigma_{\mathrm{g}}}\, \mathrm{e}^{-|\mathrm{z_{rj}^m} - \mathrm{z_{rj}^-}|/\sigma_{\mathrm{z}}}, \quad (2)$$

with range scale z_0 and discrepancy scales σ_g, σ_z; the last factor applies to RGB-D only. A candidate is retained

and, when retained, its reliability if its sample passes the gate and, for RGB-D, $\alpha_{rj} \geq \alpha_{\min}$. Each vertex keeps the at most $n_{\max}$ retained candidates of largest α_{rj} . The verified tuples $(t_r, m_{rj}, \alpha_{rj})$ form $\mathcal{O}_j$, and $\mathcal{O} = \{\mathcal{O}_j\}_j$ is passed to Stage 2. as its *observations* $(t_r, m_{rj}, \alpha_{rj}) \in \mathcal{O}_j$. A vertex with $\mathcal{O}_j \neq \emptyset$ is *observed*; unobserved vertices follow the Stage-3 deformation without a measured target.

C. Stage 2: Corrected Surface Targets

The matched measurements keep their sensor-frame sample m_{rj} is unchanged by a later revision of the world-frame trajectory . Let coordinates when the trajectory changes. Stage 2 places them with the corrected keyframe pose $\widehat{T}_r^+$ denote the corrected pose associated with record r. RGB-D records are resolved by timestamp , while LiDARrecords use the corresponding scan pose. The corrected candidate is

$$q_{rj}^+ = \left[\widehat{T}_r^+ \widetilde{m}_{rj}\right]_{1:3}.$$

, retrieved by timestamp t_r (RGB-D, interpolated if needed) or scan index (LiDAR), giving the *corrected sample*

$$\mathsf{q}_{\mathsf{rj}}^+ = \left[\widehat{\mathsf{T}}_{\mathsf{r}}^+ \widetilde{\mathsf{m}}_{\mathsf{rj}}\right]_{1:3}. \tag{3}$$

Let $\mathcal{J}_j = \{r : (t_r, m_{rj}, \alpha_{rj}) \in \mathcal{O}_j\}$. The Stage 1 reliability gives the initial target estimate

$$\bar{q}_j = \frac{\sum_{r\in\mathcal{J}_j} \alpha_{rj} q_{rj}^+}{\sum_{r\in\mathcal{J}_j} \alpha_{rj}}.$$

A median-relative test selects a consistent subset:

$$\theta_j = \max\left(\epsilon_m,\ \kappa \operatorname*{median}_{r\in\mathcal{J}_j} \|q_{rj}^+ - \bar{q}_j\|\right),$$
$$I_j = \left\{r \in \mathcal{J}_j : \|q_{rj}^+ - \bar{q}_j\| \leq \theta_j\right\}.$$

Here Let $\mathcal{J}_j$ index the observations of v_j^- whose pose exists; a vertex with $\mathcal{J}_j = \emptyset$ is treated as unobserved. Each sample is weighted by its reliability, damped with scale ζ_m by depth (RGB-D), $\omega_{rj} = \alpha_{rj}/(1 + z_{rj}^m/\zeta_m)$, or by displacement (LiDAR), $\omega_{rj} = \alpha_{rj}/(1+\|q_{rj}^+ - v_j^-\|/\zeta_m)$. Samples farther than $\theta_j = \max(\epsilon_m, \kappa \operatorname{median}_{r\in\mathcal{J}_j} \|q_{rj}^+ - \bar{q}_j\|)$ from their ω-weighted mean $\bar{q}_j$ are dropped, leaving I_j; ϵ_m reflects is a floor at the modality's sampling resolutionand κ is shared across modalities. When , $\kappa > 1$ keeps I_j nonempty, and with fewer than three candidates are available, no trimming is applied. The final samples none is dropped. The observation-derived target is

$$v_j^* = \frac{\sum_{r\in I_j} \alpha_{rj} q_{rj}^+}{\sum_{r\in I_j} \alpha_{rj}}.$$

$$\mathsf{v}_{\mathsf{j}}^* = \frac{\sum_{\mathsf{r}\in\mathsf{I}_\mathsf{j}} \omega_{\mathsf{rj}} \mathsf{q}_{\mathsf{rj}}^+}{\sum_{\mathsf{r}\in\mathsf{I}_\mathsf{j}} \omega_{\mathsf{rj}}}, \tag{4}$$

Each supported source vertex yields one target displacement $v_j^* - v_j^-$ for Stage 3. and each observed vertex has displacement $d_j = v_j^* - v_j^-$.

Algorithm 1 Stage 3 of one correction

Require: source mesh (V^-, F); measured controls d_j
Ensure: paired surfaces $\bar{\mathcal{M}}^-, \mathcal{M}^+$
label components; remove degenerate faces and control-free fragments
(a) remove outlier controls once
(b) omit fragments without controls; add synthetic controls
if $|V^-| > 1.3P_{\max}$ **then**
    (c) build proxy $\mathcal{P}^-$; (d) proxy targets, Eq. (5)
    (e) solve Eq. (6) on $\mathcal{P}^-$; (f) transfer to source
else
    (e) solve Eq. (6) on the source mesh
end if
for at most n_{rep} passes **do** ▷ (f)
    relax invalid faces; stop if none or no progress
end for
remove faces violating τ_{e} or τ_{A} and omitted components

D. Stage 3: Surface Deformation

Stage 3 converts the sparse observation-derived targets into a coherent, topology-aware surface deformation. It consolidates controls within connected components, propagates their motion through ARAP, and produces paired propagates the targets over the surface by as-rigid-as-possible (ARAP) deformation [17], which penalizes local shape distortion while fixing *controls* at prescribed positions; observed vertices give *measured controls*. Large meshes are solved on a component-aware proxy whose motion is transferred back, so the source and warped surfaces for Stage 4.

Let $d_j = v_j^* - v_j^-$ denote acontrol displacement. Before coarsening, a component-restricted inverse-distance predictor and median meshes share their triangles. Connected components follow mesh edges and are relabelled at each correction; *fragments* have fewer than N_{f} faces. Algorithm 1 orders steps (a) – MAD residual test remove statistically isolated controls without mixing disconnected surfaces.Components supported only by rejected controls are omitted. Retained components without a surviving measured control receive up to three spatially separated synthetic constraints carrying robust motion from nearby accepted controls (f).

(a) Outlier controls. The prediction $\hat{d}_j$ averages the displacements of the k nearest other controls with weights $\|v_j^- - v_i^-\|^{-2}$, using only same-component neighbours when at least $n_{\min}$ exist, so that nearby surface copies caused by drift keep separate motions. With $e_j = \|d_j - \hat{d}_j\|$, $\mu = \operatorname{median}_j e_j$, and the median absolute deviation $\mathrm{MAD} = \operatorname{median}_j |e_j - \mu|$ [30], control j is an *outlier* if $e_j > \mu + \gamma \max(1.4826\,\mathrm{MAD}, \sigma_{\min})$. These constraints complete otherwise unconstrained ARAP blocks while remaining distinct from observation-derived controls.

For large source meshes, component-aware spatial clustering [31] constructs a triangular *(b) Components without controls.* A fragment that lost all its controls is *omitted*; every other component without a control receives three *synthetic controls* at mutually distant vertices, with the coordinate-wise median displacement of the k controls nearest to its centroid.

(c) Proxy. A mesh with more than $1.3P_{\max}$ vertices is deformed through a proxy $\mathcal{P}^- = (P^-, F_P)$ built by vertex clustering [31]: each (cell, component) pair of cubic cells with side $h = (A_{\mathcal{M}}/P_{\max})^{1/2}$, $A_{\mathcal{M}}$ the mesh area, becomes one proxy vertex p_a^-, placed with the

quadric error metric (QEM) [32]; $\pi(j)$ is the proxy vertex of at most $P_{\max} \approx 60\,\mathrm{K}$ vertices. One QEMrepresentative [32]is selected per occupied cell, while different connected components remain separate during clustering. Let $\pi : V^- \to P^-$ map each source vertex to its proxy representative. Controls mapped to proxy vertex a are combined as

$$p_a^{(0)} = p_a^- + \frac{1}{|\mathcal{K}_a|} \sum_{j \in \mathcal{K}_a} d_j, \qquad \mathcal{K}_a = \{j : \pi(j) = a\}.$$

v_j^-.

The proxy targets are further consolidated by a robust local rigid field. For each controlled proxy vertex, WarpVox fits a local rigid transform to neighbouring source – target pairs using Welsch reweighting [33], [34]*(d) Proxy targets.* Vertex a starts from $p_a^{(0)} = p_a^- + |\mathcal{K}_a|^{-1} \sum_{j \in \mathcal{K}_a} d_j$, the mean displacement of its controls $\mathcal{K}_a = \{j \in \mathcal{C} : \pi(j) = a\}$ among the surviving measured and synthetic controls $\mathcal{C}$, and is *controlled* if $\mathcal{K}_a \neq \emptyset$; a proxy component without one has three vertices controlled at $p_a^{(0)} = p_a^-$. $\mathcal{M}_P$ holds the controlled vertices with a measured control; the others have *synthetic* targets. A local rigid fit $(\hat{R}_a, \hat{t}_a)$ to the pairs $(p_b^-, p_b^{(0)})$ of the k nearest $b \in \mathcal{M}_P \setminus \{a\}$, by Procrustes alignment [33] reweighted with the Welsch weights $\nu(r) = e^{-r^2/2\sigma_a^2}$ [34] of its residuals r_b at robust scale $\sigma_a = \mathrm{median}_b\, r_b + 1.4826\,\mathrm{MAD}_b(r_b)$ per round, floored at $10^{-6}\max(1\,\mathrm{m}, R_a)$ for k-th-neighbour distance R_a, predicts $\hat{p}_a = \hat{R}_a p_a^- + \hat{t}_a$. A measured target is replaced by its rigid prediction only when it is a gross residual and such residuals form *gross* if $r_a = \|p_a^{(0)} - \hat{p}_a\| \geq c_g \sigma_a$, and $2G_K > N_K$ denotes a strict majority among measured targets in its component with valid predictions. Synthetic targets are regularized by the same field; let $\mathcal{C}_P$ index the resulting targets p_a^*. of gross targets among the N_K fitted measured targets of component K. The constrained target is

$$\mathsf{p}_a^* = \begin{cases} \hat{p}_a, & a \text{ gross},\ 2G_K > N_K, \\ \hat{p}_a + \nu(r_a)(p_a^{(0)} - \hat{p}_a), & a \notin \mathcal{M}_P, \\ p_a^{(0)}, & \text{otherwise}, \end{cases} \tag{5}$$

and a vertex with too few neighbours for a fit keeps $p_a^{(0)}$.

WarpVox propagates the consolidated targets with ARAP [17]. Let $\Delta p_{ab}^{\pm} = p_a^{\pm} - p_b^{\pm}$. We solve

$$\begin{aligned} \min_{P^+, \{R_a\}} \quad & \frac{1}{2} \sum_{a=1}^{|P^-|} \sum_{b \in \mathcal{N}_P(a)} c_{ab} \\ & \cdot \left\| \Delta p_{ab}^+ - R_a \Delta p_{ab}^- \right\|^2, \\ \text{s.t.} \quad & R_a \in SO(3), \\ & p_a^+ = p_a^*, \qquad a \in \mathcal{C}_P. \end{aligned}$$

(e) ARAP. The deformed proxy positions $P^+ = \{p_a^+\}$ are fixed to p_a^* at controlled vertices $\mathcal{C}_P$ and optimized elsewhere, with one rotation $R_a \in SO(3)$ per vertex:

$$\min_{\mathsf{P}^+, \{\mathsf{R}_a\}} \sum_{\mathsf{a}} \sum_{\mathsf{f} \in \mathrm{star}_P(\mathsf{a})} \sum_{(\mathsf{b},\mathsf{c}) \in \mathsf{f}} \frac{c_{\mathsf{bc}}^{\mathsf{f}}}{3} \|(\mathsf{p}_\mathsf{b}^+ - \mathsf{p}_\mathsf{c}^+) - \mathsf{R}_\mathsf{a}(\mathsf{p}_\mathsf{b}^- - \mathsf{p}_\mathsf{c}^-)\|^2, \tag{6}$$

Here $\mathcal{N}_P(a)$ is the proxy where $\mathrm{star}_P(a)$ is the one-ring and c_{ab} are symmetric cotangent-derived weightsof proxy triangles incident to a, (b, c) runs over their edges, and $c_{bc}^f = \frac{1}{2}\cot\vartheta_{bc}^f$ uses

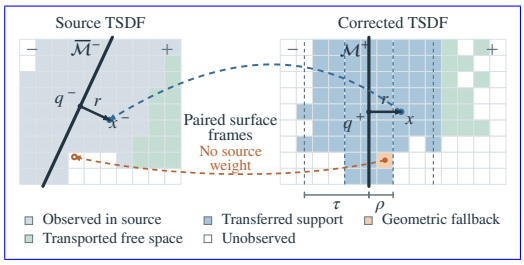


Fig. 5. Stage 4: transferred, fallback, and free-space support.

the angle opposite (b, c) in rest triangle f; summing over incident faces gives the symmetric cotangent weight $c_{bc} = \sum_{f \supset (b,c)} c_{bc}^f$ [35]. The targets provide positional constraints, while the ARAP energy propagates their motion through locally rigid surface deformation. The system is solved with libigl [36] . The standard configuration uses the proxy; the full-resolution ablation applies the same objective directly to the source mesh. Libigl [36] runs 20 local – global steps.

Proxy displacements are transferred to the source mesh by topology-local barycentric interpolation. Let *(f) Transfer and repair.* Each source vertex takes the proxy motion *topology-locally*, from the rest triangle f_j be the closest proxy triangle of $\mathrm{star}_P(\pi(j))$ closest to v_j^- within $\mathrm{star}(\pi(j))$. With barycentric coordinates λ_{jb} of its : $v_j^+ = v_j^- + \sum_{b \in f_j} \lambda_{jb}(p_b^+ - p_b^-)$, with λ_{jb} the barycentric coordinates of the closest point,

$$v_j^+ = v_j^- + \sum_{b \in f_j} \lambda_{jb}(p_b^+ - p_b^-), \qquad \lambda_{jb} \geq 0, \quad \sum_{b \in f_j} \lambda_{jb} = 1.$$

Restricting the interpolation to $\mathrm{star}(\pi(j))$ prevents motion transfer between spatially close but disconnected surfaces.

After interpolation, faces violating edge-stretch, area-change, or aspect-ratio bounds define a repair region. Incident displacements are iteratively relaxed toward mesh one-ring means with surrounding valid displacements fixed. Faces still violating edge-stretch so the transfer stays within its component. Repair flags a face as *invalid* if its edge stretch $\xi^{\mathrm{e}} = \max_e \ell_e^+/\ell_e^- \geq \tau_{\mathrm{e}}$, area growth $\xi^{\mathrm{A}} = A^+/A^- \geq \tau_{\mathrm{A}}$, or area-change bounds are removed from both surfaces. No new triangles or connections are introduced, so

$$\bar{\mathcal{M}}^- = (\bar{V}^-, \bar{F}), \qquad \mathcal{M}^+ = (\bar{V}^+, \bar{F}), \qquad \bar{F} \subseteq F.$$

The shared face indexing provides the source – warped correspondence used in Stage 4. aspect ratio $\xi^{\mathrm{asp}} = (\ell_{\max}^+)^2/(2A^+) \geq \tau_{\mathrm{asp}}$ with $\ell_{\max}^+ \geq \ell_{\min}$, for source and warped edge lengths $\ell^{\pm}$ and areas $A^{\pm}$; Jacobi relaxation moves the displacement of each of its vertices halfway to the mean of its edge neighbours, all others fixed, until the largest update falls below 10^{-5} m. The retained triangles $\bar{F} \subseteq F$ define the paired surfaces $\bar{\mathcal{M}}^- = (\bar{V}^-, \bar{F})$ and $\mathcal{M}^+ = (\bar{V}^+, \bar{F})$.

E. Stage 4: Source-Supported TSDF Reconstruction

Stage 4 reconstructs a native TSDF from the paired source and warped surfaces produced by Stage 3. The warped surface defines the corrected zero set , while restores the volumetric state for continued fusion at the source resolution: $\mathcal{M}^+$ provides the zero set and distance magnitudes, and the source TSDF provides volumetric state that a surface alone does not encode: observed support , sign orientation, integration weights , and free space. Distance magnitude is recomputed

from the warped surface, whereas source state is transferred through the paired surface correspondence(Φ^-, w^-) supplies observed support and integration weights and guides the sign.

Let (Φ^-, w^-) denote the source TSDF. The corrected volume uses the same voxel size s and truncation distance *Surface-local reconstruction* covers voxel centres x within τ as the source volume. We write $\rho = 3s/2$ for the near-surface margin. WarpVox first allocates the blocks containing warped-surface vertices together with one layer of neighbouring blocks. Surface-local reconstruction is restricted to this neighbourhood, while transported free space may subsequently allocate additional blocks.

For each voxel x in the surface-local domain, a of $\mathcal{M}^+$ in blocks with a warped vertex and their 26 neighbours. A closest-triangle query [37] on $\mathcal{M}^+$ returns face $f$returns face $f = (a, b, c)$, closest point q^+, and barycentric coordinates β. The shared face indices give the corresponding source point , and $q^- = \sum_{k \in f} \beta_k \bar{v}_k^-$.

Let $R_f^\pm \in SO(3)$ be paired local frames built from the same indexed triangle edge and corresponding source and warped mesh normals. Their relative rotation defines on $\bar{\mathcal{M}}^-$. With unit normals $n_f^\pm$ of f on $\bar{\mathcal{M}}^-$ and $\mathcal{M}^+$, oriented by the vertex order of F, and unit directions $t_f^\pm$ of edge (a, b), frames $R_f^\pm = [t_f^\pm, n_f^\pm \times t_f^\pm, n_f^\pm] \in SO(3)$ define the local-rigid pullback

$$\Psi_f(x) = q^- + R_f^-(R_f^+)^\top(x - q^+).$$

$$\Psi_{\mathsf{f}}(\mathsf{x}) = \mathsf{q}^- + \mathsf{R}_{\mathsf{f}}^-(\mathsf{R}_{\mathsf{f}}^+)^\top(\mathsf{x} - \mathsf{q}^+), \tag{7}$$

This map preserves tangential and normal offsets relative to the paired triangleswhich maps the surface-relative offset into the source triangle's frame [38]; absolute TSDF sign is determined separately from the source field. Let

$$x^- = \Psi_f(x), \quad r(x) = \|x - q^+\|, \quad d^+(x) = \min(\tau, r(x)).$$

let $x^- = \Psi_f(x)$, $r(x) = \|x - q^+\|$, and $d^+(x) = \min(\tau, r(x))$. A trilinear source sample is valid only when at x^- is *valid* if every source voxel with a nonzero interpolation coefficient has positive weight; its interpolated weight is then transferred. If complete interpolation is unavailable but the source voxel containing x^- has positive weight, that weight provides support without copying its distance value. If neither form of source support exists, coefficient is supported, and *sign-consistent* if their distances share one nonzero sign. The transferred weight, capped at $w_{\max}$, is

$$\mathsf{w}_{\mathrm{src}}^+(\mathsf{x}) = \begin{cases} \bar{w}^-(x^-), & \text{valid sample at } x^-, \\ w^-(y_0(x^-)), & \text{otherwise,} \end{cases} \tag{8}$$

with interpolated weight $\bar{w}^-$ and containing voxel $y_0(x^-)$; x enters the fallback branch. Denote the transferred weight by $w_{\mathrm{src}}^+(x)$.

The warped mesh normal induces the is *source-supported* if $w_{\mathrm{src}}^+(x) > 0$. With geometric sign $s_f^+(x) = \mathrm{sgn}((x - q^+)^\top n_f^+)$. When source TSDF samples are available on both sides of the paired source , the source TSDF sets the sign

$$\mathsf{s}_{\mathrm{tr}}(\mathsf{x}) = \begin{cases} s_f^+(x)\,\mathrm{sgn}(g_f), & g_f \text{ available,} \\ \mathrm{sgn}\,\bar{\Phi}^-(x^-), & \text{else, if sign-consistent,} \\ s_f^+(x), & \text{otherwise,} \end{cases} \tag{9}$$

where $\bar{\Phi}^-$ is the interpolated source distance, and $g_f = [\bar{\Phi}^-(q^- + s\,n_f^-) - \bar{\Phi}^-(q^- - s\,n_f^-)]/(2s)$, its central difference across the paired source face, their central difference determines whether this sign is retained or flipped. Otherwise, a consistent source sample is used when available, with $s_f^+(x)$ as fallback. Denote the result by $s_{\mathrm{tr}}(x)$calibrates the normal toward free space, available if all three samples are valid and $g_f \neq 0$.

Valid source samples *Transferred* voxels are source-supported: narrow set $\mathcal{N}_{\mathrm{tr}}$ with $r(x) \le \rho$, and extended set $\mathcal{E}$ with $r(x) \le \rho$ form narrow transferred support; those with $\rho < r(x) \le \tau$ form candidate extended support. Both receive

$$(\Phi_{\mathrm{tr}}^+(x), w_{\mathrm{tr}}^+(x)) = (s_{\mathrm{tr}}(x)d^+(x), w_{\mathrm{src}}^+(x)).$$

$\rho < r(x) \le \tau$, which keeps a voxel with a valid sample only if g_f is available.

Where transferred source support is unavailable, voxels satisfying $r(x) \le \rho$ receive a finite surface prior , unless their closest surface feature is an open boundary:

$$w_{\mathrm{fb}}(x) = \mathrm{clamp}_{[0,1]}\left(\frac{\rho - r(x)}{s}\right).$$

This prior provides compact supportfor continued fusion and is not interpreted as recovered observation count. Its supported voxels *Fallback* supplies a surface prior where the pullback has no source support, $w_{\mathrm{src}}^+(x) = 0$ and $r(x) \le \rho$, with weight $w_{\mathrm{fb}}(x) = \mathrm{clamp}_{[0,1]}((\rho - r(x))/s)$, withheld where q^+ lies on a boundary edge (incident to one face) or its endpoint; the voxels with $w_{\mathrm{fb}}(x) > 0$ form Ω_{fb}.

Candidate extended support is filtered on the output voxel grid. Marching Cubes reads sign changes along grid edges, which join face-adjacent voxels , so a candidate with an opposite-sign positive-weight face (6-)neighbour is removed. A flood fill over face-connected candidates starts from those touching narrow transferred or fallback support and ends when no new candidate is reached. The narrow samples and reached candidates After removing the voxels of $\mathcal{E}$ that would add a zero crossing or are disconnected from $\mathcal{N}_{\mathrm{tr}} \cup \Omega_{\mathrm{fb}}$, $\mathcal{N}_{\mathrm{tr}}$ and the rest of $\mathcal{E}$ form Ω_{tr}.

Surface-local reconstruction does not recover observed free space farther from the corrected zero set*Observed-volume transport* carries the source state through the same paired frames. Observed source voxels satisfying $w^-(y) > 0$ and $\Phi^-(y) \ge \rho$ are eligible for forward transport. Within each source TSDF block, they are binned into axis-aligned cellsof $\mathrm{round}(\tau/s)$ voxels per side; the mean voxel-centre position represents each nonempty group. Group membership is purely spatial and does not require voxel connectivity.

For group $y \in \Omega^-$ are grouped into truncation-sized cells; each cell c , the closest valid source face uses the source face closest to its observed-voxel centroid, unless it or its warped copy is degenerate, and the paired warped face to its representative supplies q_c^- and barycentric coordinates; the same face index and coordinates supply q_c^+ on the warped surface. The corresponding triangle-frame construction gives $R_c^\pm$. A group is discarded when no admissible face match is found or either paired frame is degenerate. The forward map is

$$\Gamma_c(y) = q_c^+ + R_c^+(R_c^-)^\top(y - q_c^-).$$

Each $\Gamma_c(y)$ is assigned to its containing output voxel. Insertion is fill-only: existing positive-weight targets are not overwritten. Because any two voxels of one Marching-Cubes cell lie in a common 26-neighbourhood, targets in the 26-neighbourhood of retained negative support are rejected, so transported free space cannot enable a new sign-crossing cell. Accepted samples retain their source-derived positive values and weights, defining Ω_{fs} and $(\Phi_{\mathrm{fs}}^+, w_{\mathrm{fs}}^+)$. define $\Gamma_c(y) = q_c^+ + R_c^+(R_c^-)^\top(y - q_c^-)$, and output voxels are sampled at $\Gamma_c^{-1}(x)$. Samples that reach one output voxel from connected equal or

26-neighbouring source voxels form one sheet, which contributes the distance and weight of its most interior (L_∞) sample; distinct sheets are fused as in Voxblox, by weighted mean distance and summed weight. On the reconstructed band, the sheet touching the source voxel of the pullback of x is already represented, and the others add their weight $w^+_{\mathrm{add}}(x)$. Elsewhere, sheets fill empty voxels: fills of opposite sign to a reconstructed 26-neighbour are dropped, and a minimum s–t cut keeps the maximum-weight fills without opposite-sign neighbours, so no zero crossing is added, forming Ω_{vol} with Φ^+_{vol} and w^+_{vol}.

Because fallback applies only without transferred support and free-space transport writes only unsupported voxels The three sets are disjoint, Ω_{tr}, Ω_{fb}, and Ω_{fs} are disjoint. With $\Omega^+ = \Omega_{\mathrm{tr}} \cup \Omega_{\mathrm{fb}} \cup \Omega_{\mathrm{fs}}$, the corrected field is

$$(\Phi^+(x), w^+(x)) = \begin{cases} (\Phi^+_{\mathrm{tr}}(x), w^+_{\mathrm{tr}}(x)), & x \in \Omega_{\mathrm{tr}}, \\ (s^+_f(x) d^+(x), w_{\mathrm{fb}}(x)), & x \in \Omega_{\mathrm{fb}}, \\ (\Phi^+_{\mathrm{fs}}(x), w^+_{\mathrm{fs}}(x)), & x \in \Omega_{\mathrm{fs}}. \end{cases}$$

$\Omega^+ = \Omega_{\mathrm{tr}} \cup \Omega_{\mathrm{fb}} \cup \Omega_{\mathrm{vol}}$ is the set of supported output voxels:

$$(\Phi^+, \widetilde{w}^+) = \begin{cases} (s_{\mathrm{tr}} d^+,\ w^+_{\mathrm{src}} + w^+_{\mathrm{add}}), & x \in \Omega_{\mathrm{tr}}, \\ (s^+_f d^+,\ w_{\mathrm{fb}} + w^+_{\mathrm{add}}), & x \in \Omega_{\mathrm{fb}}, \\ (\Phi^+_{\mathrm{vol}},\ w^+_{\mathrm{vol}}), & x \in \Omega_{\mathrm{vol}}, \end{cases} \tag{10}$$

Voxels outside Ω^+ have zero weight. When the source TSDF is unavailable, reconstruction reduces to the finite surface prior. The resulting volume is serialized as a with $w^+ = \min(w_{\max}, \widetilde{w}^+)$. The resulting native Voxblox TSDF and replaces the active map. Its zero set follows the corrected geometry, available observation state is inherited from the source volume, and subsequent fusion and later corrections operate directly on the updated TSDF, and fusion continues on it.

F. Complexity

Let V_q be the number of queried source vertices, Q the number of returned surface–keyframe candidates, $K_{\mathrm{KF}} = |\mathcal{C}_{\mathrm{KF}}|$ the number of cached keyframes , $A = |\mathcal{A}|$ the number of active source cells , $P = |P^-|$ the that of examined vertex – keyframe pairs, K_{KF} that of stored keyframes with $|D|$ samples each, A that of Marching-Cubes cells that emit triangles of $\mathcal{M}^-$, P that of vertices in the deformation solve (the proxy size, or $|V^-|$ on the direct path), and X^-, X^+ the source /and output voxel slotsin Stage . A correction costs $C_{\mathrm{corr}} = C_{\mathrm{assoc}} + C_{\mathrm{def}}(A, P) + C_{\mathrm{vol}}(X^-, X^+)$ over Stages 1 – 2, 3, and 4. The per-correction cost is

$$C_{\mathrm{corr}} = C_{\mathrm{assoc}} + C_{\mathrm{def}}(A, P) + C_{\mathrm{vol}}(X^-, X^+).$$

Sensor-position indices are maintained online rather than rebuilt at correction time, giving $C_{\mathrm{assoc}} = O(V_q \log K_{\mathrm{KF}} + Q)$ Reading every keyframe and rebuilding the index gives $C_{\mathrm{assoc}} = O(K_{\mathrm{KF}}|D| + (K_{\mathrm{KF}}+V_q)\log K_{\mathrm{KF}}+Q)$ for RGB-D KD-tree queries and $O(V_q+Q)$ for LiDARgrid queries. At fixed resolution, $V_q = O(A)$. With $\bar{k} = Q/V_q$, session length affects association through K_{KF} and $\bar{k}$, whereas map size affects deformation and reconstruction through A, X^-, and X^+. By construction, $P \le P_{\max}$, and $O(K_{\mathrm{KF}}|D| + V_q + Q)$ for LiDAR; C_{def} covers proxy construction, target consolidation, ARAP, transfer, and repair, and C_{vol} surface-local reconstruction and observed-volume transport. Reintegrating N frames costs $\Theta(N|D|)$.

Full reintegration costs $C_{\mathrm{reint}} = \Theta(\sum_{n=1}^{N} |D_n|) = \Theta(N|D|)$. When $K_{\mathrm{KF}} = \Theta(N)$, both retain the same frame-count order, but WarpVox avoids integrating the $|D|$ range samples from every historical frame.

TABLE II

MEAN RECONSTRUCTION ACCURACY AND CORE TIME AT 0.25 M.

Data Method F@10 (%) F@25 (%) F@50 (%) Time (s) ↓ Uncorrected 7.00 19.29 34.91 – Open3D 8.13 22.94 41.30 – Voxgraph 8.27 22.78 39.49 5.54 Wavemap 1.73 11.54 30.91 8.32 Voxblox reint. 10.93 28.71 48.67 109.25 **WarpVox** 11.13 28.76 48.39 2.26 Uncorrected 7.13 27.91 49.22 – Open3D 13.50 38.58 61.02 – Voxgraph 9.38 31.89 52.49 4.53 Wavemap 5.23 33.10 62.55 2.68 Voxblox reint. 13.21 44.29 67.45 15.73 **WarpVox** 13.00 43.97 67.07 4.87

Data	Method	F@10 (%) ↑	F@25 (%) ↑	F@50 (%) ↑	CD (cm) ↓	Time (s) ↓
Kimera (39)	Uncorrected	7.00	19.29	34.91	247.7	–
	Open3D	8.13	22.94	41.30	114.9	–
	Voxgraph	8.27	22.78	39.49	142.2	5.54
	Wavemap	1.73	11.54	30.91	132.1	8.32
	Voxblox reint.	10.93	28.71	48.67	101.9	109.25
	WarpVox	11.13	28.76	48.39	103.1	2.26
KITTI (12)	Uncorrected	7.13	27.91	49.22	126.2	–
	Open3D	13.50	38.58	61.02	82.9	–
	Voxgraph	9.38	31.89	52.49	124.8	4.53
	Wavemap	5.23	33.10	62.55	71.1	2.68
	Voxblox reint.	13.21	44.29	67.45	67.3	15.73
	WarpVox	13.00	43.97	67.07	68.8	4.87

A dashmarks an untimed update: the uncorrected map applies no correction , and Open3D was (Uncorrected) or not timed (Open3D).

IV. EXPERIMENTS

A. Evaluation Protocol

Data and execution. Each bag is processed once end-to-end; loop-closure checkpoints are not replayed. ORB-SLAM3 and Voxblox run from the start, and each WarpVox correction replaces the active volume before fusion resumes. Keyframe range images are cached online, eliminating bag rewind and separate preprocessing at loop events. Associations, targets, and deformation state are discarded after each correction, while the cache and corrected TSDF persist. We evaluate 39 RGB-D loop closures from 18 Kimera-Multi sequences [39] and 12 LiDAR cases from six KITTI sequences [40]. The PGMO comparison, repeated correction, and ablation share ten Kimera-Multi sequences, and the 0.10 m evaluation uses 11 of the 39 cases; both subsets were sampled once. The uncompressed 16-bit RGB-D cache occupies 0.9 –3.6– 3.6 GiB (mean 2.1 GiB; 1349 – 5707 keyframes).

Reference and alignment. All methods reconstruct the same temporal interval. Kimera-Multi uses the official ground-truth point cloud cropped to a 5 m tube around the GT ground-truth (GT) trajectory, matching Voxblox's maximum integration range. KITTI uses the Velodyne cloud aggregated at ground-truth poses within the mapper's 25 m ray range. Predictions outside the GT frame are aligned using only timestamp-matched trajectory positions; no mesh-to-GT registration is used. Voxgraph uses the trajectory recovered from of its optimized submap-pose graph, while reconstructions driven by the corrected ORB-SLAM3 [6] trajectory share one alignment per case.

Baselines. Uncorrected Voxblox uses the pre-correction trajectory. Open3D [41] provides an independent TSDF rebuild, Wavemap [18] a wavelet occupancy rebuild, and Voxblox

TABLE III

MEAN WARPVOX CORE-TIME BREAKDOWN IN SECONDS.

Setting	Assoc.	Target	ARAP	Topol.	TSDF	Total ↓
Kimera, 25 cm (39)	0.199	0.399	1.419	0.008	0.231	2.256
KITTI, 25 cm (12)	0.310	0.069	2.457	0.147	1.888	4.872
Kimera, 10 cm (11)	2.623	0.244	5.298	0.004	1.038	9.207

TABLE IV

PGMO CORRECTED-SURFACE COMPARISON ON TEN BAGS.PGMO corrected-surface comparison on ten sequences.

Method	F@10 (%) ↑	F@25 (%) ↑	F@50 (%) ↑	CD (cm) ↓	Time (s) ↓
Kimera-VIO pre-mesh	4.680	14.094	26.277	309.6	–
Kimera-PGMO one-shot	8.639	23.875	42.318	179.0	8.231
WarpVox Stage 1 – 3	8.440	24.134	43.231	152.9	3.515

reintegration [4] the representation-matched TSDF reference. These methods re-fuse every range frame using the corrected trajectory interpolated to each timestamp. Voxgraph [11] optimizes rigid TSDF submaps in its independently optimized frame. WarpVox starts from the active uncorrected TSDF and uses keyframe measurements only for target construction. own frame.

Metrics and timing**.** We report F@10/25/50 from bidirectional nearest-neighbour tests at 0.10/0.25/0.50 m and the symmetric Chamfer distance (CD), the mean nearest-neighbour distance in both directions. All methods share each case's crop, two-million-point sample, random seed, and thresholdsand random seed. Main TSDF experiments use 0.25 m voxels and 1.0 m truncation; Wavemap uses a matched 0.25 m finest level and is evaluated on its official occupancy point cloud. The 11-case high-resolution subset and ten-bag subset were each sampled once; the latter is shared by the PGMO comparison and ablation.

Timing. We report in-memory core Core map-update time is measured in memory on an AMD Ryzen 7 9700X with 64 GB RAM , including TSDF or occupancy integration, submap or mesh and covers integration, optimization, or the applicable WarpVox stages; , excluding disk I/O and PGMO MeshFrontend constructionare excluded.

B. Reconstruction Fidelity and Update Cost

At 0.25 m, WarpVox is the closest method to reintegration at every threshold (Table II).

On Kimera-Multi, WarpVox differs from reintegration by $+0.20/+0.05/-0.28$ pp at F@10/F@25/F@50. 50, and on KITTI by -0.32 pp at F@25 in dataset mean. Its CD stays within 1.5 cm of reintegration on both datasets, below every other correction baseline. Per-event F@25 differences fall on both sides of zero, from -2.91 to $+2.59$ pp on Kimera-Multi and from -3.03 to $+1.80$ pp on KITTI. WarpVox is $48.43\times$ faster than reintegration on Kimera-Multi and $3.23\times$ on KITTI. The smaller KITTI gain is consistent with KITTI speedup reflects lower-cost sparse-LiDAR replay and WarpVox's larger active surface and output volume. Voxgraph improves over uncorrected Voxblox at every threshold by aligning rigid SDF submaps, but stays 5.98 and 12.08 pp below WarpVox at F@25 on the two datasets, consistent with uncorrected intra-submap drift. Wavemap shows low F@10 but competitive KITTI F@50. This is 50, consistent with a hierarchical occupancy representation whose 0.25 m setting is its finest level rather than a uniform map resolution; its finest occupancy level whose grid-centred evaluation points can also sit points lie up to 0.22 m from the true surface.

At 0.10 m on 11 Kimera-Multi cases, WarpVox obtains 17.13/39.77/57.83 at F@10/25/50 versus 17.21/39.68/57.12 for reintegration, while reducing core time from 192.548 to 9.207 s ($20.91\times$). Table III separates association, target estimation, deformation, topology processing, and TSDF reconstruction at both resolutions. Stage shares follow their workloads: LiDAR association searches spherical-image neighbourhoods, and RGB-D target construction aggregates more retained observations. Figure 6 complements the quantitative comparison at 0.25 m.

C. Comparison with *Kimera-PGMO*Deformation-Based Correction

This experiment compares Kimera-PGMO's deformation-graph correction with WarpVox's observation-derived targets andproxy ARAP. Both use , like WarpVox, propagates trajectory corrections to a fused map, and, like ElasticFusion, deforms the surface through an embedded deformation graph, which ties each mesh vertex to the pose at its timestamp; WarpVox ties each surface region to the keyframes that observe it. Both correct the same timestamped 8 cm Kimera-VIO pre-mesh and with the same VIO pre-correction and PGMO-optimized trajectories; PGMO uses the , PGMO through its official `mesh_trajectory_deformer` tool [9] , whereas WarpVox applies and WarpVox through Stages 1 – 3. 3, isolating the correction method.

WarpVox improves F@10 (%) F25(%) /F@50 (%)Time (s) ↓ Kimera-VIO pre-mesh 4.680 14.094 26.277 – Kimera-PGMO by $0.26/0.91$ pp and lowers CD from 179.0 cm for one-shot 8.639 23.875 42.318 8.231 WarpVox Stage 1 – 3 8.440 24.134 43.231 3.515

PGMO, the stronger of its two modes (183.8 cm online), to 152.9 cm, while reducing corrected-surface core time by $2.34\times$. At F@10, which approaches the 8 cm frontend-mesh resolution, WarpVox it remains within 0.20 pp of Kimera-PGMO. It improves Stage 4 then reconstructs an 8 cm native TSDF in 1.277 s, with F@25/F@50 by 0.26/0.91 pp while reducing corrected-surface core time by $2.34\times$. WarpVox subsequently applies Stage 4 to reconstruct an 8 cm native TSDF , adding 1.277 s (4.792 s for Stages 1 – 4)and yielding Fof $24.387/43.098\%$; this complete update (4.792 s) is faster than PGMO's surface correction alone (8.231 s) and stays above it at F@25/F@50 scores of 24.387%/43.098%. 50.

ElasticFusion [12] deforms its surfel map through an embedded deformation graph driven by its own tracking. On TUM [42] fr3/long_office_household, its own room-scale benchmark, it tracks at 1.80 cm absolute trajectory error (ATE), with 62.77% F@2 and 3.69 cm CD; WarpVox reaches 72.13% and 2.82 cm. On the larger Kimera-Multi sequences, its tracking accumulates a mean ATE of 115.8 m, leaving its map at 1.98% F@25. WarpVox applies trajectory corrections to TSDF maps

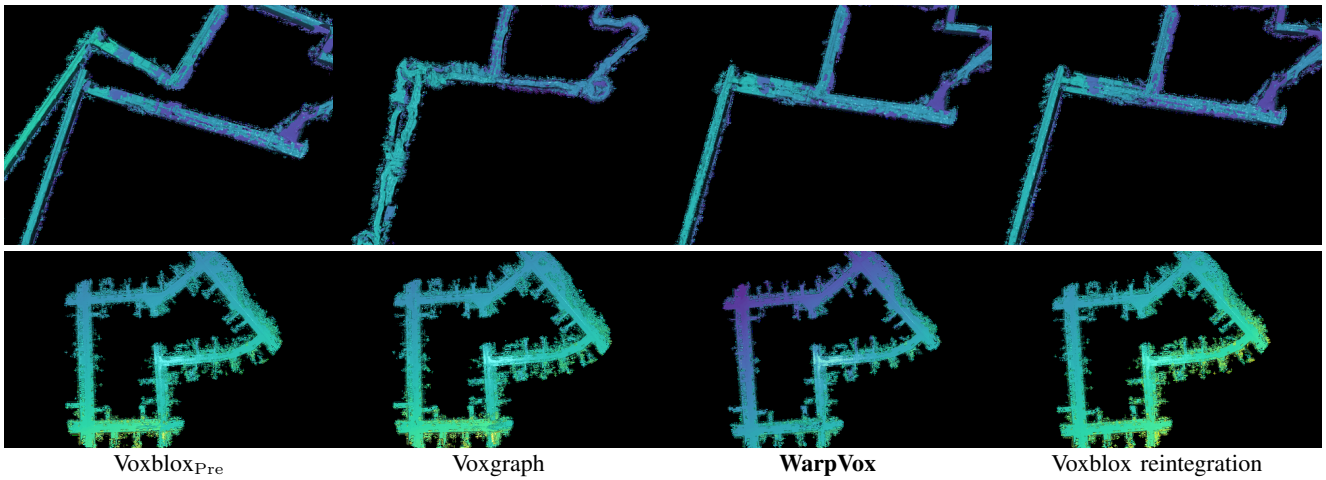


Fig. 6. **Qualitative reconstruction comparison at 0.25 m.** Four One Kimera-Multi loop-closure cases are shown in the case (topfour rows) and two one KITTI cases in the case (bottomtwo rows;), each row uses in the same view across methods. Colors encode vertex height.

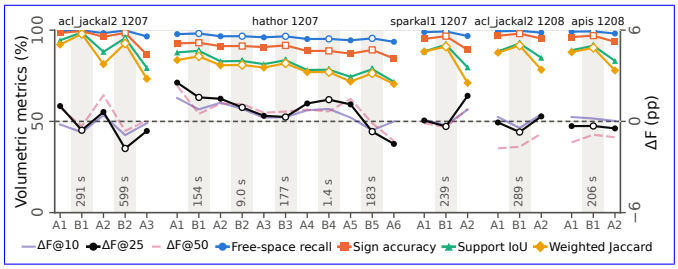


Fig. 7. **Repeated correction (A, 15) and continued fusion (B, 10).**

TABLE V

ABLATION OF STAGES 1 – 4 ON TEN KIMERA-MULTI CASES.

Variant	F@10 (%) ↑	F@25 (%) ↑	F@50 (%) ↑	S1 – 2 (s) ↓	S3 (s) ↓	S4 (s) ↓
Full-mesh ARAP	10.278	26.749	45.490	–	3.323	–
Robust 60K QEM proxy	10.277	26.717	45.436	–	1.408	–
Nearest-KF pose transport	9.872	25.979	44.546	0.044	–	–
Mesh-only TSDF	9.982	26.400	45.068	–	–	0.098
Source-supported TSDF (Final)	10.155	26.616	45.392	0.480	1.408	0.160

Times are listed only for the stage under comparison. The upper block varies S3, stops before Stage 4, and is scored on the ARAP surface; the lower varies S1 – 2 and S4 against Final and is scored on meshes re-extracted from the output TSDF.

for RGB-D and LiDAR alike, which suits the growing scale and sensing of today's mapping data.

D. TSDF Consistency over Repeated Correction

We follow the five multi-loop sequences of the ten through repeated correction and continued fusion, with checkpoints after each correction (A) and after continued fusion up to the next loop closure (B; Fig. 7). F-scores compare each map with the ground truth. The volumetric metrics compare it with Voxblox reintegration of the same measurement history, as the reference since no ground-truth TSDF exists. On a shared grid with unallocated voxels at zero weight, they are support IoU over positive-weight voxels, weighted Jaccard similarity $\sum \min(w, w_{\mathrm{r}}) / \sum \max(w, w_{\mathrm{r}})$, w_{r}-weighted free-space recall over voxels with $\Phi_{\mathrm{r}} > 0.5s$, and sign accuracy on common support outside the $0.5s$ zero-level band.

At F@25, WarpVox differs from reintegration by $+0.50$ pp after correction and -0.01 pp after continued fusion on average, with per-event differences from -1.48 to $+2.55$ pp after correction. Over the 15 post-correction checkpoints, it averages 83.4%, 80.1%, 97.3%, and 92.2% on the four volumetric metrics. Rebuilding the TSDF from the corrected mesh at each correction differs from reintegration mainly in the extent of the shared support measured by IoU and weighted Jaccard, while free-space recall and sign accuracy remain high. Continued fusion with the same Voxblox update raises all four metrics in every interval, restoring agreement in revisited regions.

E. Ablation Study

Table V isolates one design choice at a time on one loop-closure case from each of the ten bagssequences, at 0.25 m.

Nearest-KF moves each control by its closest keyframe's pose correction. Observation-derived targets aim to reverse fusion: they recover the keyframe measurements that formed the source TSDF at each vertex, then screen, weight, and combine them (Eq. (4)). They improve all three F-scores over this baseline, including 0.637 pp at F@25, most of it from the observations identified in Stage 1. Replacing only their measured samples of Eq. (3) with each keyframe's pose transport leaves WarpVox 0.042 pp ahead at F@25 on average and in 8 of 10 cases (paired: $+0.07$, -0.04, $+0.08$, $+0.05$, $+0.05$, $+0.02$, $+0.07$, -0.01, $+0.04$, $+0.09$ pp), while removing the gate and reliability screening drops F@10/25/50 to 8.93/25.49/45.29%, below nearest-keyframe transport at F25. The robust 60K QEM proxy remains within 0.032 pp of full-mesh ARAP at F@25 while reducing Stage 3 time from 3.323 to 1.408 s. Stage 4 keeps the re-extracted mesh within 0.122 pp of the robust Stage 3 surface and improves F@25 by 0.216 pp over mesh-only conversion, at an additional cost of 0.062 s. Varying the association, trimming,

TABLE VI

GAUSSIAN CASE STUDY ON ONE KIMERA-MULTI SEQUENCE.

Method	F@25 (%) ↑	F@50 (%) ↑	Depth L1 (cm) ↓	Time ↓
SplaTAM pre-correction	17.23	35.25	173.93	–
WarpVox	23.12	40.59	142.54	21 s
SplaTAM corrected-rebuild	23.25	41.01	141.37	≈ 2 h

and Stage-3 thresholds one at a time changes mean F@25 by -0.19 to $+0.05$ pp, and the observation cap $n_{\max} \in \{1, 4, 8, 15\}$ by at most 0.019 pp.

Beyond the stage ablation, reintegrating only the cached keyframes on the same ten cases reaches $24.18/41.62$ at F@25/F@50 in 33.70 s and still takes , over an order of magnitude longer than WarpVox, while remaining $2.44/3.78$ pp below it. The cache therefore supports target construction but not a complete volumetric rebuild.

TSDF consistency. We compare WarpVox with corrected-pose Voxblox reintegration on a shared voxel grid, treating unallocated voxels as zero weight. We report positive-weight support IoU, weighted Jaccard similarity of voxel weights, reference-weighted free-space support recall for $\Phi_r > 0.5s$, and TSDF-sign accuracy on common support outside the $0.5s$ zero-level band. Averaged over ten 0.25-m checkpoints, WarpVox achieves 86.19%, 80.06%, 98.50%, and 96.29%, respectively, showing substantial agreement in support, accumulated weight, and TSDF sign structure.

Cross-representation 3DGS case study. Table VI evaluates whether the observation-derived correction transfers beyond TSDFs. We apply the same observation-derived targets to a SplaTAM [26] reconstruction based on 3D Gaussian Splatting [27]. This is a geometry-only diagnostic: each Gaussian centre is displaced by the resulting ARAP deformation, while colours, opacities, scales, covariances, densification, pruning, and rendering losses all appearance and optimization parameters remain fixed. For comparison, we retain the original pre-correction SplaTAM reconstruction and rebuild SplaTAM from scratch using the corrected trajectory.

The centre-only correction approaches the corrected-pose rebuild at both surface thresholds and in depth error, while taking 21 s instead of approximately 2 h. With appearance parameters fixed, PSNR and MS-SSIM increase from 11.96 to 12.59 and from 0.512 to 0.526, while LPIPS decreases from 0.430 to 0.411. This single-sequence case study tests target transferability only; WarpVox's implementation and primary evaluation remain focused on native TSDF correction for continued robotic mappingThe same targets thus also correct a Gaussian map.

V. LIMITATIONS

WarpVox assumes a static scene, valid geometry, and a locally coherent pre-correction TSDF. It cannot recover geometry absent from the source mapor repair incorrectly fused topology, and repeated or nearly overlapping surfaces may still produce an internally consistent but incorrect association. Keyframe-cache memory WarpVox targets static scenes and corrects the geometry already in the map: later observations enter through continued fusion, and severe trajectory failures remain recoverable by reintegration from the original measurements. The keyframe cache grows with session length. Finite-resolution reconstruction cannot exactly match the reintegrated map. Topology processing may remove a few faces with unresolved deformation-induced distortion. Residual trajectory error limits accuracy; severe failures require reintegration from the original measurements. , at 2.1 GiB on average for RGB-D. Beyond the large-scale benchmarks evaluated here, future work covers dynamic scenes, cache compaction, and longer sessions with more loop closures.

VI. CONCLUSION

WarpVox corrects an active TSDF after loop closure without replaying the complete frame history. Across RGB-D and LiDAR benchmarks, it remains close to full reintegration and reaches with a $48\times$ mean speedup on RGB-D, and it corrects the surface more accurately than Kimera-PGMO in less than half the time. The resulting native Voxblox volume resumes fusion after a seconds-scale update , and supports later corrections, and retains only a keyframe range cache for historical measurements. The Gaussian case study suggests that observation-derived targets may extend beyond voxel grids; code the same targets also correct a Gaussian map. Code and configurations will be released.

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